

Latent Manipulation of 2D Brain MRI using Volume-Preserving GANs

- Develop a GAN model capable of manipulating the latent space to simulate age-related changes in brain MRI.
- Prior to training, apply preprocessing that preserves the original brain volume, ensuring that anatomical structures remain accurate and realistic. By systematically modifying the age attribute in the latent vector, investigate how brain structures evolve over time, while keeping other anatomical features unchanged to isolate the effect of aging.

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Simulating brain aging with generative models

Aging leads to structural changes:

- Cortical thinning
- Ventricular enlargement

GANs can learn data distributions and generate realistic images



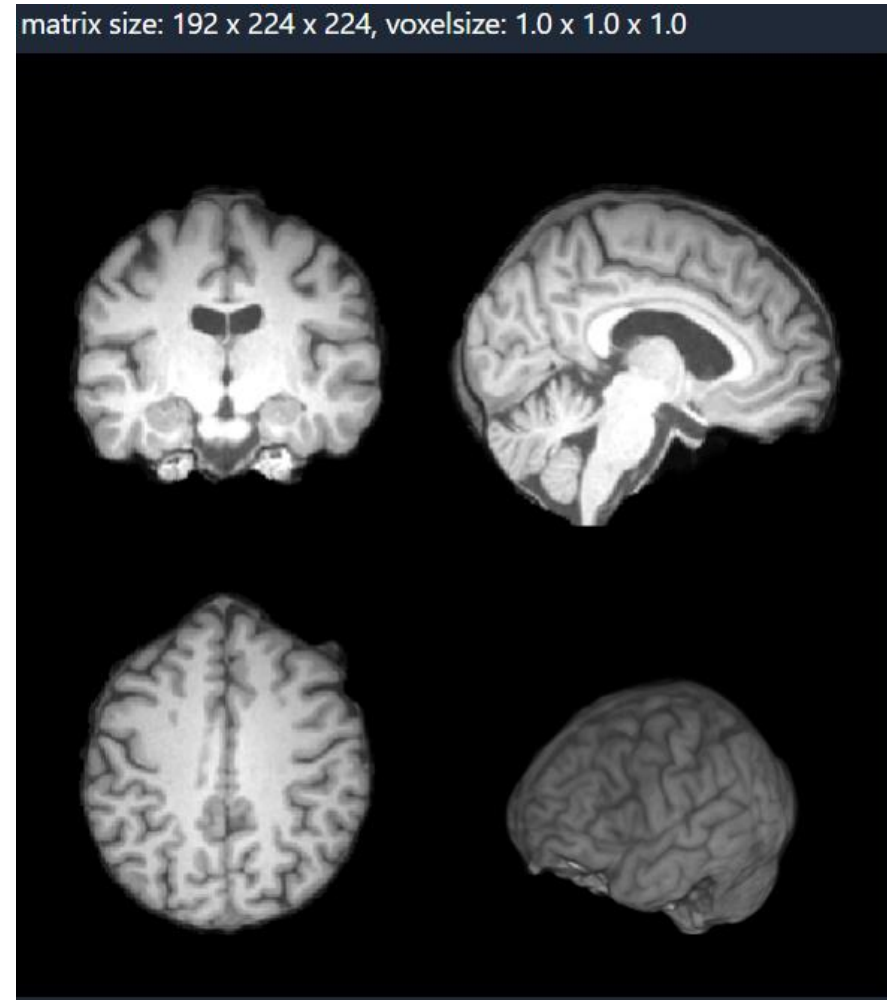
Problem – Preprocessing Effect

Wrong brain scale



GM 815ml

Good brain scale

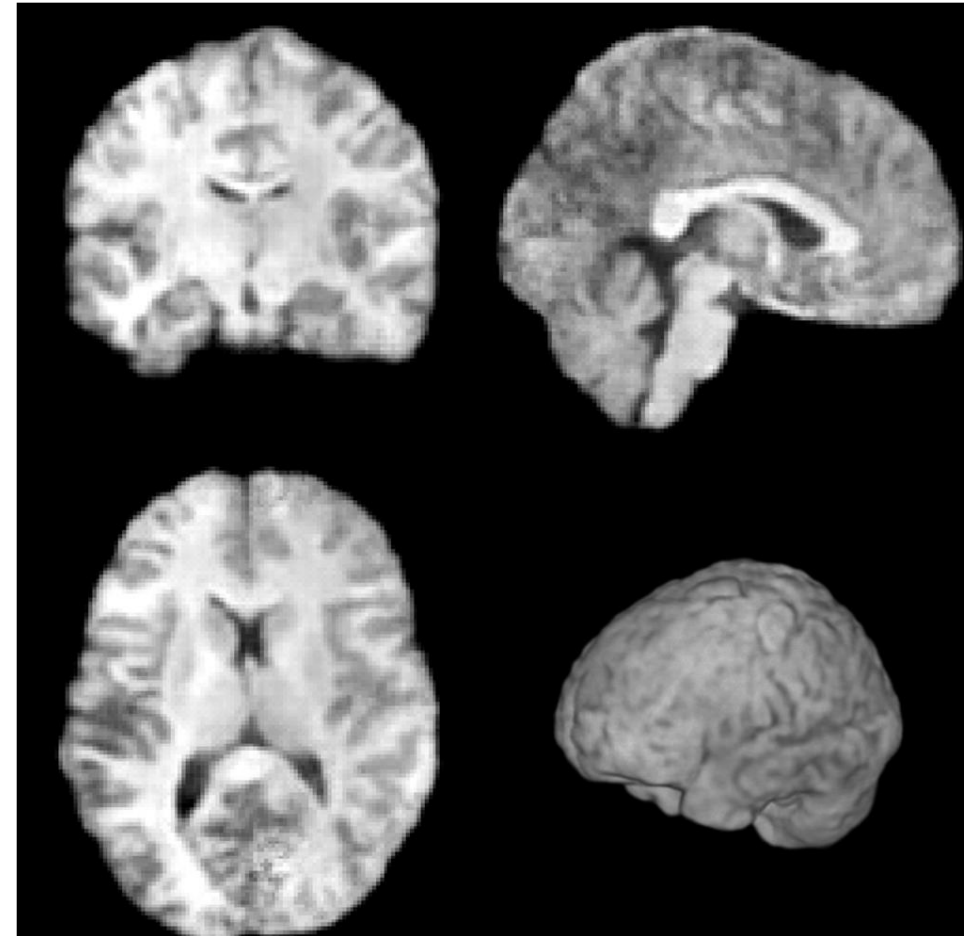


GM 620ml

Problem – GAN Impact

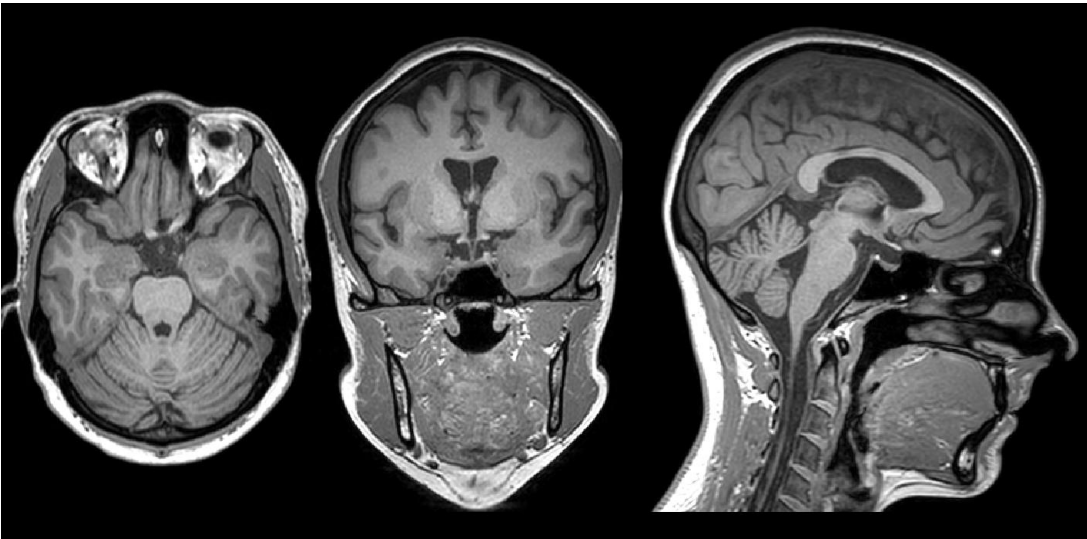
High Variance: GAN must learn both anatomical features AND absolute head size differences.

Convergence: Slower training; model often produces **blurry results** due to poor spatial overlap of structures.



Method - Preprocessing

Dataset: >5000 Raw open-source MRI



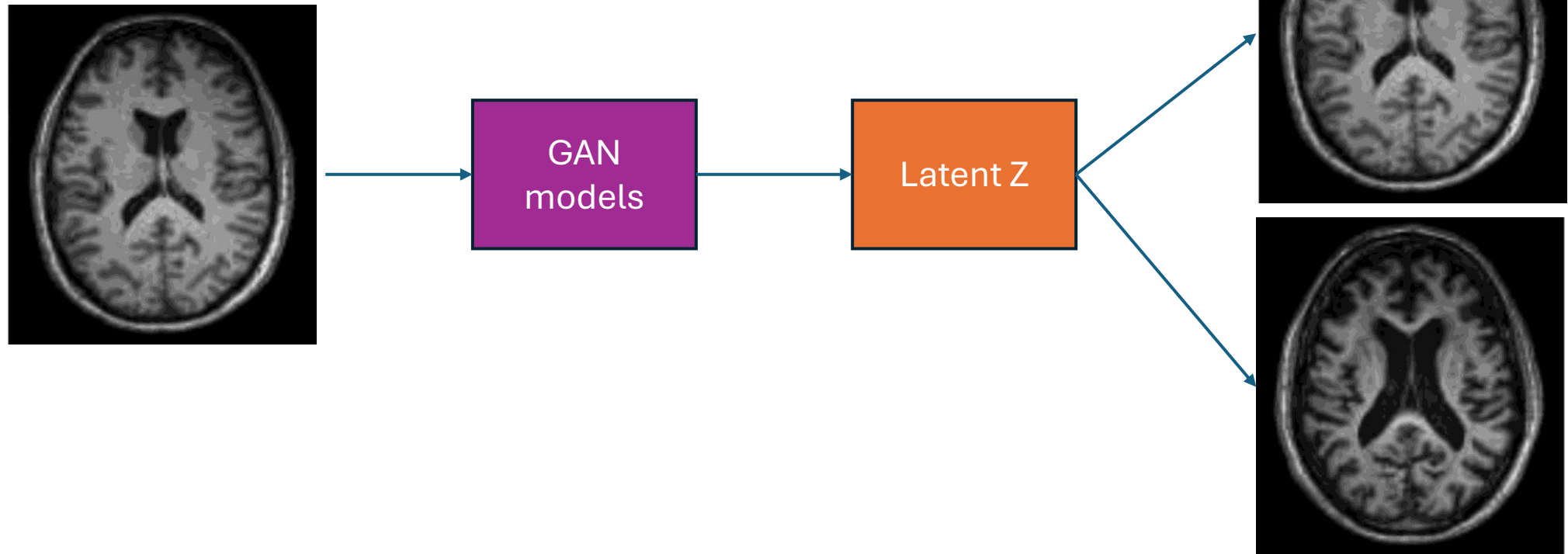
Preprocessing step: Conform, Skull-stripping, Alignment, ...

Henschel et al., 2020



Method – GAN Training

- Preserve original brain volume during preprocessing
- Learn **age-aware latent space**
- Enable controlled manipulation:
 - Same anatomy, different ages



Method – Volume Measurement

